Project: Worked examples
Title: Bolts and fillet welds (EN 1993-1-8)
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1. Scope and method

The design resistances of one bolt in a plate (shear, bearing, tension, punching, combined shear and tension), an optional block-tearing check of the bolt group, the spacing limits, and a fillet weld by the simplified and the directional method, each against its design force.

WHAT IT CALCULATES	RELATION
Bolt strengths	f_{ub} = class × 100 MPa; $A_s = \pi/4\cdot((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$
Shear (cl. 3.6, Table 3.4)	$F_{v,Rd} = \alpha_v \cdot f_{ub} \cdot A \cdot n_s/\gamma_{M2}$; α_v = 0.6 (4.6, 5.6, 8.8 or shank in the plane), 0.5 (4.8, 5.8, 6.8, 10.9 with threads in the plane)
Bolt bearing resistance	$F_{b,Rd} = k_h \cdot \min(k_1 \cdot \alpha_b \cdot f_u \cdot d \cdot t/\gamma_{M2}$, [single lap, one bolt row: $1.5 \cdot f_u \cdot d \cdot t/\gamma_{M2}$]); $\alpha_b = \min(\alpha_d, f_{ub}/f_u, 1)$, α_d = $e_1/3d_o$ (end) or $p_1/3d_o - 1/4$ (inner); $k_1 = \min(2.8e_2/d_o - 1.7, 1.4p_2/d_o - 1.7, 2.5)$; k_h = 1 (normal holes), 0.8 (oversize), 0.6 (slot across the load)
Bolt tension resistance	$F_{t,Rd} = k_2 \cdot f_{ub} \cdot A_s/\gamma_{M2}$, k_2 = 0.9; punching $B_{p,Rd} = 0.6\pi \cdot d_m \cdot t \cdot f_u/\gamma_{M2}$, $d_m = (s + e)/2$ from the across-flats s and the across-corners e of the smaller of head and nut, or d_m = s when e is not entered (conservative); combined $F_v/F_{v,Rd} + F_t/(1.4F_{t,Rd}) \le 1$
Block tearing (cl. 3.10.2)	$V_{eff,1,Rd} = f_u \cdot A_{nt}/\gamma_{M2} + f_y \cdot A_{nv}/(\sqrt{3} \cdot \gamma_{M0})$ for a group loaded concentrically; $V_{eff,2,Rd} = 0.5 \cdot f_u \cdot A_{nt}/\gamma_{M2} + f_y \cdot A_{nv}/(\sqrt{3} \cdot \gamma_{M0})$ for an eccentrically loaded group
Weld, simplified (cl. 4.5.3.3)	$f_{vw,d} = f_u/(\sqrt{3} \cdot \beta_w \cdot \gamma_{M2})$, $F_{w,Rd} = f_{vw,d} \cdot a$; directional (cl. 4.5.3.2): $\sqrt{(\sigma^2 + 3(\tau^2 + \tau^2))} \le f_u/(\beta_w \cdot \gamma_{M2})$, $\sigma \le 0.9f_u/\gamma_{M2}$

Basis: Positioning of holes for bolts[1](cl. 3.5, Table 3.3); Design resistance of individual fasteners[1](cl. 3.6.1, Table 3.4); Block tearing[1](cl. 3.10.2); Fillet welds: directional and simplified methods; correlation factor[1](cl. 4.5.3.2, 4.5.3.3, Table 4.1); Nominal stress area (formula)[2](cl. 9.1).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
grade	Bolt property class	8.8		drawing: SPL-001 rev B	any
size	Bolt size	M20		drawing: SPL-001 rev B	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$hole_{type}$	Hole type: oversize holes reduce the bearing resistance by 0.8, a slot across the load by 0.6 (a d_o above the normal clearance is treated as oversize)	Normal clearance		default	any
$single_{lap}$	Single-lap joint with only one bolt row (limits the bearing resistance to $1.5 \cdot f_u \cdot d \cdot t / \gamma_{M2}$, cl. 3.6.1)	no		default	any
n_s	Shear planes	2		drawing: SPL-001 rev B	1 … 4
threads	Threads in the shear plane	yes		assumption: Threads may lie in the shear planes (short bolts).	any
steel	Plate steel grade (β_w)	S275		drawing: SPL-001 rev B	any
t	Plate thickness	10	mm	drawing: SPL-001 rev B	3 … 50 mm
f_y	Plate yield strength	275	MPa	spec: EN 10025-2 S275, t ≤ 16 mm	235 … 460 MPa
f_u	Plate ultimate strength	410	MPa	spec: EN 10025-2 S275, 3 ≤ t ≤ 100 mm	340 … 720 MPa
e_1	End distance (load direction)	40	mm	drawing: SPL-001 rev B	10 … 500 mm
e_2	Edge distance (across the load)	35	mm	drawing: SPL-001 rev B	10 … 500 mm
p_1	Pitch (load direction; blank: one row)	70	mm	drawing: SPL-001 rev B	20 … 1000 mm
k_{e1}	Minimum e_1/d_o	1.2		spec: EN 1993-1-8 Table 3.3	1 … 3
k_{e2}	Minimum e_2/d_o	1.2		spec: EN 1993-1-8 Table 3.3	1 … 3
k_{p1}	Minimum p_1/d_o	2.2		spec: EN 1993-1-8 Table 3.3	2 … 4
k_{p2}	Minimum p_2/d_o	2.4		spec: EN 1993-1-8 Table 3.3	2 … 4
γ_{M2}	γ_{M2}	1.25		spec: UK NA to BS EN 1993-1-8, NA.2.3	1.1 … 1.5
γ_{M0}	γ_{M0}	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.2

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$F_{v,Ed}$	Shear per bolt	55	kN	assumption: Governing bolt from the bolt-group calculation (M3).	0 … 2000 kN
$F_{t,Ed}$	Tension per bolt	0	kN	assumption: No tension.	0 … 2000 kN
$block_{load}$	Block tearing loading: concentric ($V_{eff,1}$) or eccentric ($V_{eff,2}$, the tension term halved)	Concentric loading (V_eff,1)		default	any

3. Assumptions

A1. Static loading; category A (bearing type) shear and category D (non-preloaded) tension; no slip-resistant joints or moment end plates. (Scope of S4.)
A2. Bearing is taken at the worst of an end bolt and an inner bolt, and at an edge bolt across the load (k_1 with e_2 and p_2). (Conservative for every bolt in the group.)
A3. Oversize holes reduce the bearing resistance by 0.8 and a slot across the load by 0.6; a d_o above the normal clearance (d + 2, d + 3 from M27) with the hole type left at normal is treated as oversize. The single-lap cap on the bearing resistance applies only when the single-lap switch is on (a warning says when it would bind). (EN 1993-1-8 cl. 3.6.1, Table 3.4.)
A4. Spacing minima (Table 3.3) and γ_M factors are entered from a National Annex record; the values shown default to EN 1993-1-8 recommended. (A small set of factors only; no table from a standard is shipped.)
A5. Group forces come from M3 and weld forces per length from M6 by link. (Group forces and weld forces are linked from the bolt-group and weld-group calculations.)
A6. threads (Threads in the shear plane) is taken as yes. (Threads may lie in the shear planes (short bolts).)
A7. $F_{v,Ed}$ (Shear per bolt) is taken as 55 kN. (Governing bolt from the bolt-group calculation (M3).)
A8. $F_{t,Ed}$ (Tension per bolt) is taken as 0 kN. (No tension.)

4. Working

[20] f_{ub} = class first figure × 100 [1](Table 3.1)
= class 8.8
= 800 MPa
Bolt ultimate strength

[21] $f_{yb} = f_{ub}$ × second figure / 10 [1](Table 3.1)
= class 8.8
= 640 MPa
Bolt yield strength

[22] $A_s = \pi/4 \cdot ((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495P$, $d_3 = d - 1.2269P$ [2](cl. 9.1)
= d = 20, P = 2.5 mm
= 244.79 mm²
Tensile stress area

[23] d_0 = d + 2 (≤ M24), d + 3 (≥ M27) unless entered
= 22 mm
= 22 mm
Hole diameter

[24] $\alpha_v = \alpha_v$ (threads in the shear plane) [1](Table 3.4)
= class 8.8
= 0.6
Shear factor

[25] $F_{v,Rd} = \alpha_v \cdot f_{ub} \cdot A \cdot n_s / \gamma_{M2}$ [1](Table 3.4)
$= 0.6 \times 800 \times 244.79 \times 2 / 1.25$
$= 188\ kN$
Shear resistance per bolt
[26] $\alpha_b = \min(\alpha_d, f_{ub}/f_u, 1)$, $\alpha_d = \min(e_1/3d_0, p_1/3d_0 - 1/4)$ [1](Table 3.4)
$= \min(0.60606, 0.81061, 1.9512, 1)$
$= 0.60606$
Bearing factor α_b
[27] $k_1 = \min(2.8e_2/d_0 - 1.7, 1.4p_2/d_0 - 1.7, 2.5)$ [1](Table 3.4)
$= e_2 = 35,\ p_2 = —,\ d_0 = 22$
$= 2.5$
Bearing factor k_1
[29] $F_{b,Rd} = k_1 \cdot \alpha_b \cdot f_u \cdot d \cdot t / \gamma_{M2}$ [1](cl. 3.6.1, Table 3.4)
$= 2.5 \times 0.60606 \times 410 \times 20 \times 10 / 1.25$
$= 99.394\ kN$
Bearing resistance per bolt
[30] $F_{t,Rd} = k_2 \cdot f_{ub} \cdot A_s / \gamma_{M2}$, $k_2 = 0.9$ [1](Table 3.4)
$= 0.9 \times 800 \times 244.79 / 1.25$
$= 141\ kN$
Tension resistance per bolt
[31] $B_{p,Rd} = 0.6\pi \cdot d_m \cdot t \cdot f_u / \gamma_{M2}$, $d_m = s$ (e not entered) [1](Table 3.4)
$= 0.6\pi \times 30 \times 10 \times 410 / 1.25$
$= 185.48\ kN$
Punching shear resistance
[32] $u_v = F_{v,Ed} / F_{v,Rd}$
$= 55\ kN / 188\ kN$
$= 0.29255$
Shear utilisation
[33] $u_b = F_{v,Ed} / F_{b,Rd}$
$= 55\ kN / 99.394\ kN$
$= 0.55335$
Bearing utilisation
[34] $u_t = F_{t,Ed} / \min(F_{t,Rd}, B_{p,Rd})$
$= 0\ kN / \min(141\ kN, 185.48\ kN)$
$= 0$
Tension utilisation (tension or punching)
[35] $u_{vt} = F_{v,Ed} / F_{v,Rd} + F_{t,Ed} / (1.4 * F_{t,Rd})$ [1](Table 3.4)
$= 55\ kN / 188\ kN + 0\ kN / (1.4 \times 141\ kN)$
$= 0.29255$
Combined shear and tension

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{v,Rd}$	188.00 kN	Shear resistance
$F_{b,Rd}$	99.39 kN	Bearing resistance
$F_{t,Rd}$	141.00 kN	Tension resistance
A_s	244.8 mm²	Stress area
f_{ub}	800 MPa	Bolt ultimate strength
f_{yb}	640 MPa	Bolt yield strength
α_v	0.6	Shear factor

SYMBOL	VALUE	DESCRIPTION
α_b	0.60606	Bearing factor
k_1	2.5	Bearing factor k_1
$B_{p,Rd}$	185.48 kN	Punching resistance
governing	"bearing (0.553)"	Governing bolt mode
u_v	0.29	Shear utilisation
u_b	0.55	Bearing utilisation (governing)
u_t	0.00	Tension utilisation
u_{vt}	0.29	Combined utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bolt bearing	u_b = 0.55	$u_b \le 1$	0.55	[1](Table 3.4)
PASS	Bolt shear	u_v = 0.29	$u_v \le 1$	0.29	[1](Table 3.4)
PASS	Combined shear and tension	u_{vt} = 0.29	$u_{vt} \le 1$	0.29	[1](Table 3.4)
PASS	Bolt tension and punching	u_t = 0.00	$u_t \le 1$	0.00	[1](Table 3.4)
PASS	Spacing and edge distances (Table 3.3 minima)	e_1 40, e_2 35, p_1 70, p_2 – mm	≥ 26.4, 26.4, 48.4, 52.8 mm	–	[1](Table 3.3)

7. Applicability

WARNING records: tabulated values not taken from a reference record are γ_{M2}, γ_{M0}, k_{e1}, k_{e2}, k_{p1}, k_{p2} (EN 1993-1-8 National Annex factors); f_y, f_u (Plate strengths). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
$hole_{type}$: not entered, so the default of 'Normal clearance' is used.
$block_{load}$: not entered, so the default of 'Concentric loading ($V_{eff,1}$)' is used.

8. Verification

```
Template struct.connection-ec3@1.7.0 Content hash e5ff72f057654ae1 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. International Organization for Standardization. Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel—Part 1: Bolts, Screws and Studs With Specified Property Classes—Coarse Thread and Fine Pitch Thread. ISO 898-1:2013. ISO; 2013.